

HCSL: Rumor Detection by Integrating Intra-Sample Curriculum Learning and Hierarchical Semantic Learning

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Abstract

The proliferation of misinformation on social media is compromising the credibility of digital content and jeopardizing the security and stability of the online information ecosystem. Most of the existing rumor detection methods struggle to model multi-granular semantic information on deep propagation structures and alleviate noise interference encountered during deep graph training. In this work, we propose HCSL, a novel rumor detection framework that combines intra-sample curriculum learning (ISCL) and hierarchical semantic learning (HSL). ISCL adaptively selects the graph nodes to obtain a more stable optimization path for rumor representation by considering the node depth in the propagation tree. HSL employs two lightweight BERT encoders and a hierarchical contrastive learning to learn word-level and post-level representations, respectively. Furthermore, a semantic alignment mechanism is used to capture user-tendency features and propagation-structure features consistently. Extensive experiments are conducted on two public rumor detection datasets. Our experimental results demonstrate that HCSL significantly outperforms several baseline methods.

CCS Concepts

• Information systems → Data mining.

Keywords

Rumor Detection, Intra-sample Curriculum Learning, Hierarchical Semantic Learning

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1 Introduction

The rapid growth of web technologies and social media has fundamentally shaped how information is accessed and exchanged. However, the open and accessible nature of the internet has simultaneously facilitated the rampant spread of online rumors. Unverified rumors often spread rapidly online at minimal cost, disrupting everyday public discourse and potentially triggering severe social panic and instability across areas such as politics, the economy, and public health. The proliferation of rumors not only severely pollutes the online information environment but also undermines public trust. Therefore, efficient rumor detection technologies have become urgent, which mitigate the adverse societal impacts of misinformation on social media platforms.

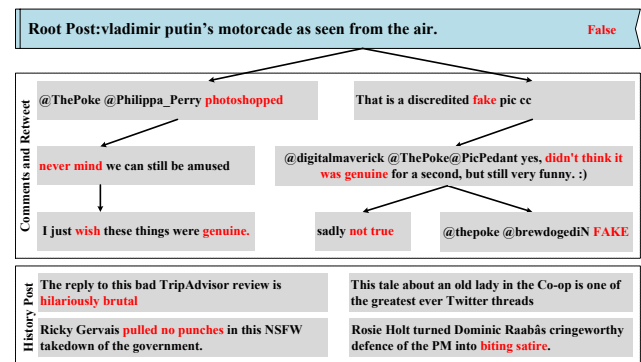


Figure 1: The top shows the source post and its related comments or retweets, while the bottom depicts the corresponding user's historical posts.

Currently, rumor detection research has shifted from traditional machine learning with manual feature selection to deep learning that autonomously extracts key features [40]. Deep learning models possess powerful representational capabilities that allow them to automatically extract deep semantic features from raw text. This process enhances the accuracy and robustness of rumor-detection tasks

[41]. Recent advancements in rumor detection typically leverage source posts from social platforms and multidimensional contextual information, including user preferences [8] and social behavioral characteristics [34], to predict whether an event is a rumor. As shown in Figure 1, the associated comments and retweets, and the user's posting history provide evidence that a consistent pattern of sharing satirical and critical content reveals that the event is a rumor. However, social media texts are often characterized by their brevity, nonstandard syntax, and implicit semantics, and the massive number of ambiguous, emotional, or even irrelevant comments and retweets derives significant noise [30]. To address these issues, numerous studies have proposed enhanced methods from different perspectives. One category of research focuses on the original post's content, using deep neural networks to represent the text features [1, 14, 18]. Another category of research leverages comments, retweets, and their structural information to extract structural features of rumor propagation by using Graph Convolutional Network (GCN) [3]. Furthermore, some approaches have attempted to incorporate user profiles and historical posting records to characterize users' behavioral patterns and credibility. Despite the significant achievements in these areas, current approaches still have two limitations. First, regarding the issue of excessive noise in the rumor-propagation graph, most curriculum-learning algorithms rank difficulty at the inter-sample level, which commence with low-difficulty data and gradually transition to more complex samples by reordering the training sample sequence. In fact, such an overly strict "from easy to difficult" strategy may be counterproductive. Although many complex samples are inherently more difficult to learn, they are indispensable for capturing the underlying patterns in data, and their informational value may significantly outweigh that of a vast quantity of simple samples [43]. Secondly, current methods usually use a single text encoder or simple fusion strategies to extract text features. Consequently, it remains challenging to fully exploit complementary information in multi-perspective text representations.

To address these limitations, we propose a novel method HCSL to detect rumor by integrating intra-sample curriculum learning and hierarchical semantic learning. First, we design a propagation tree-based intra-sample curriculum learning strategy to reduce the interference of noisy data on model training and fully exploit the information contained in the propagation structure. Inspired by [31], we construct an intra-sample difficulty metric based on the depth and topological features of different rumor-propagation substructures. This strategy aims to guide the model to learn from relatively simple parts within a sample and then progressively handle more complex and noisy data. By prioritizing reliable components with minimal noise interference, our method effectively reduces the risk of converging to local minima. Then, inspired by [4], we design a hierarchical semantic learning module to obtain more robust and discriminative text representations. We use a word-level Transformer and a post-level Transformer for feature extraction [41]. Specifically, we employ two text encoders with complementary modeling capabilities to learn representations of the source post and its related texts from different perspectives. Finally, to obtain as comprehensive information as possible, we use multi-head attention to fuse representations from each layer across different encoders into a comprehensive representation, which reduces the

susceptibility of individual encoders to noise and bias. The main contributions are as follows:

- We propose a novel rumor-detection method HCSL by integrating intra-sample curriculum learning and hierarchical semantic learning, which reduces noise interference in the data and improves the quality of post-text content extraction.
- The intra-sample curriculum learning strategy is designed to suppress noise interference in propagation graphs. And a hierarchical semantic learning module is employed to enhance the textual representation. The semantic alignment method is proposed to learn rumor representations that incorporate both rich content semantics and propagation topology information.
- Extensive experiments on the public datasets demonstrate that our proposed method HCSL obtains better effectiveness compared to the baseline methods.

2 Problem Setting

The rumor collection is formally defined as $D = \{A_1, A_2, \dots, A_n\}$, where A_i represents the i -th rumor event and n denotes the total number of events. For each event A_i , we characterize it as a triplet $A_i = \{S_i, G_i, P_i\}$, where S_i refers to the source post, G_i denotes the propagation graph structure of this post, and P_i consists of historical posts published by the source poster. An undirected graph $G_i = (V_i, E_i)$ is constructed based on the relationships between the source post S_i and its comments and retweets. Within this propagation graph, the post S_i serves as the root node, while other nodes in the node set V_i consist of interactive posts such as comments and retweets. The edge set E_i is formed by interactive behaviors between posts, such as replies and quotes. The historical post set $P_i \in \mathbb{R}^{r \times s}$, where r represents the number of historical posts, and s denotes the length of each post.

Each event A_i has a ground-truth label $y_i \in Y$. The objective of this task is to train a rumor classifier $f: A_i \rightarrow y_i$ in a supervised learning manner.

3 Methodology

3.1 Framework Overview

The architecture of HCSL is shown in Figure 2. The model extracts features in two ways: first, it employs ISCL and a GCN to capture the structural features of the rumor propagation graph; second, it uses HSL to extract features from the historical posts published by the user of the source post. Next, we will elaborate on the process of rumor classification using HCSL, which comprises several components: Feature Extraction, Intra-sample Curriculum Learning, Hierarchical Semantic Learning, Rumor Classification, and Loss Function.

3.2 Feature Extraction

Since rumor propagation graphs and user tendencies provide crucial insights, the model encoder is structured into two distinct components to capture these features. The first component encodes rumor propagation patterns by modeling dissemination mechanisms using interaction features extracted from posts. The second component

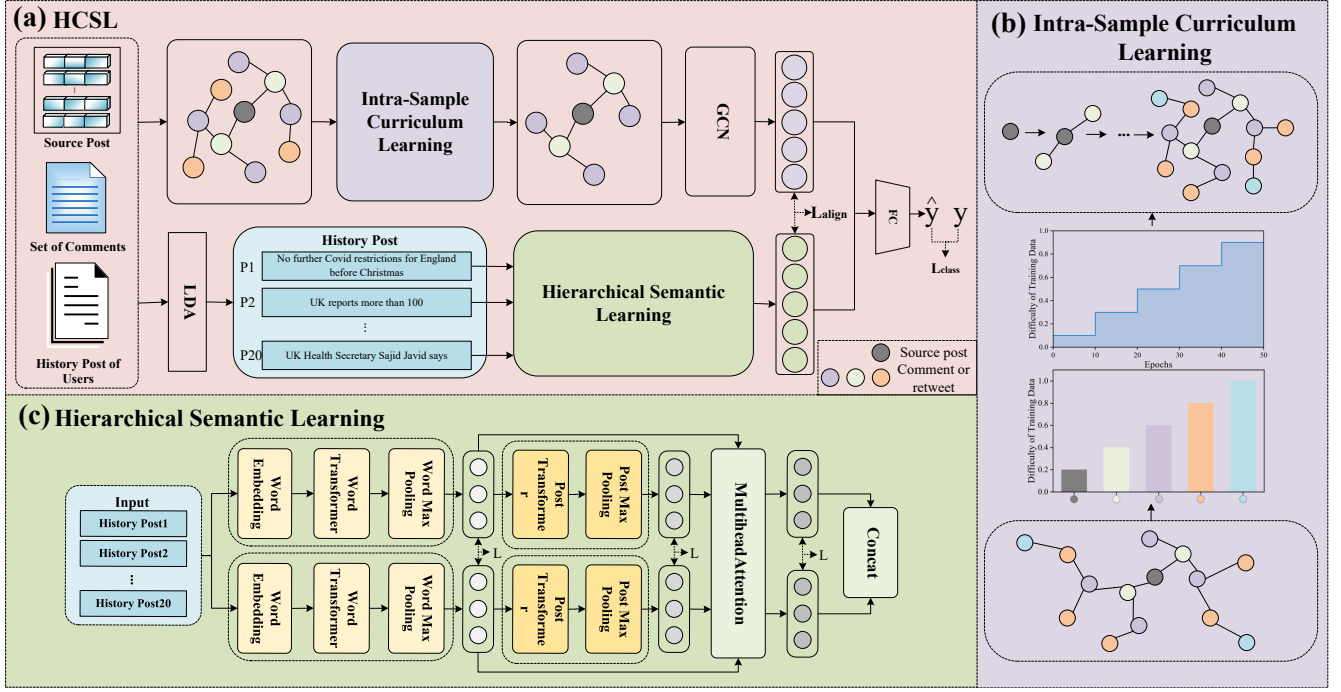


Figure 2: (a) The Architecture of HCSL. (b) Intra-sample Curriculum Learning Strategy. (c) Hierarchical Semantic Learning.

encodes users' propagation tendencies by leveraging their historical posts related to event topics. We construct an undirected graph to represent the propagation of social interaction within an event. Nodes in the graph represent posts, while edges represent interaction behaviors between posts (e.g., comments or shares). This graph is utilized for subsequent intra-sample curriculum learning and GCN-based graph structure extraction.

Let $F_i = (V'_i, E'_i)$ denote the new rumor propagation graph generated by processing the rumor propagation graph G_i through ISCL, where for $v \in V'_i$, its corresponding node feature vector $X_v \in \mathbb{R}^g$, with the feature dimension g . V'_i consists of the nodes generated by the subgraph from ISCL, and E'_i consists of the edges of the subgraph. Considering l -th layer GCN, the feature vector of node v at layer l can be expressed as:

$$a_v^{(l)} = \text{ReLU} \left(W \cdot \phi \left(\left\{ a_u^{(l-1)}, \forall u \in N(v) \cup \{v\} \right\} \right) \right),$$

where W denotes the parameter matrix. The feature transformation comprises two steps: element-wise mean pooling ϕ and the ReLU activation function, $a_u^{(l-1)}$ is the feature vector of node u at layer $l-1$, where u is a neighbor node of node v , and $\{v\}$ denotes the set containing only node v itself. $N(v)$ represents the set of nodes adjacent to node v . We initialize $a_v^{(0)} = X_v$. Subsequently, we aggregate the node features of the l -th layer to obtain the global representation h_i^G :

$$h_i^G = \text{READOUT} \left(\left\{ a_v^{(l)} \mid v \in V_i \right\} \right), \quad (1)$$

To accurately extract event-source texts via HSL, we employed the LDA topic model [15] to filter historical posts by topic, thus constructing a novel historical post dataset P_i for users who published

the source post as the training foundation. Due to limited sequence length, we cannot process all posts within a single sequence. Therefore, we independently encode each post, representing it as a matrix $T_i = \{t_1, t_2, \dots, t_m\} \in \mathbb{R}^{r \times s}$ as the input of the model, where t_i denotes the user's i -th post, r represents the number of posts, and s denotes the number of words in each post. We use two lightweight BERT encoders to perform word-level encoding and post-level encoding on the user's historical post matrix T_i , respectively. Each encoder comprises two Transformer-based encoder architectures (word-level and post-level), generating word-level contextual representations w_i^1, w_i^2 , as formulated in Eq.4 and post-level contextual representations p_i^1, p_i^2 . The word-level and post-level features from both encoders are then processed through an attention mechanism to obtain M_{atten} and N_{atten} , as formulated in Eq.5 and Eq.6. Finally, both features are concatenated to form a global text representation:

$$h_i^T = \text{CONCAT}(M_{atten}, N_{atten}), \quad (2)$$

Furthermore, we concatenate the representation h_i^G learned from the diffusion graph structure with the global text representation h_i^T to obtain the final latent embedding representation of the event:

$$h_i = \text{CONCAT}(h_i^G, h_i^T), \quad (3)$$

3.3 Intra-sample Curriculum Learning

Intra-sample curriculum learning initially focuses on learning the relatively "easier" features within each sample, and then gradually transitions to learning more complex parts within each sample in the training process. Specifically, a curriculum is defined during training to determine which subgraph of the propagation graph should be utilized at any given time point (Figure 3(b)). The module

contains two parts: One is to design difficulty metrics to identify simple and complex components. The other is to dynamically manage these transitions during training based on a curriculum learning training set.

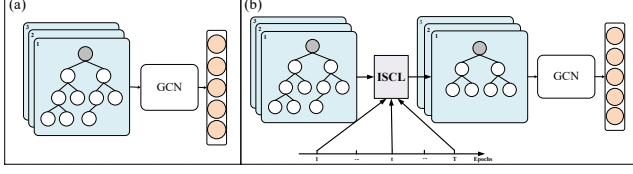


Figure 3: (a) Traditional graph structural feature extraction. (b) Graph feature extraction integrated with ISCL: as the number of epochs increases, the depth of the propagation graph structure in the training set also increases, gradually incorporating complex components and noise from the propagation graph.

3.3.1 Difficulty Measurement. We designed a difficulty coefficient based on the graph propagation depth. Shallow-level graphs have fewer nodes, shorter propagation paths, and simpler structures. In contrast, deep-level graphs have more nodes, longer paths, and more complex structures. This strategy simulates the learning process from simple to complex components within a sample. Specifically, given a rumor propagation graph G_i , where $v_{\text{root}} \in V$ denotes the source tweet node. For any node, its depth $d'(v_i)$ is defined as the shortest path length from the root node to that node. We define the difficulty of training samples as the maximum depth of the graph. During the initial phase of curriculum learning, we prune the graph to retain only nodes with depth $d'(v_i) \leq Q_c$ and their associated edges, where Q_c is the maximum depth permitted at the current curriculum stage. Formally, at curriculum stage i , the subgraph $F_i = \{V'_i, E'_i\}$ used for training is defined as:

$$V'_i = \{v_i \in V \mid \text{BFS}(v_{\text{root}}, v_i) \leq Q_c\}$$

$$E'_i = \{(u, v) \in E \mid u \in V'_i \wedge v \in V'_i\}$$

As training progresses, Q_c gradually increases until it encompasses the full depth of the original graph. It can guide the model to transition from learning local, direct propagation patterns to capturing global propagation patterns through long-range dependencies.

3.3.2 Training Schedule. To dynamically coordinate the training state of the model with sample difficulty, we designed an adaptive step scheduler based on performance feedback. Unlike traditional linear growth strategies based on a fixed number of iterations, our scheduler determines when to proceed to the next training phase based on the model's performance feedback. Specifically, the propagation graph depth Q_c is primarily determined by two metrics: performance plateau and performance stabilization. Performance plateau is the condition in which the model's accuracy fails to exceed the historical best value achieved at that depth over five consecutive training epochs, indicating a bottleneck at that depth and triggering deepening. Second, performance stabilization is the standard deviation of accuracy across the most recent three training rounds. If the value falls below 0.005, it indicates that the model's

performance has stabilized and also triggers deepening. When the above conditions are satisfied, the current allowed maximum propagation depth Q_c is calculated based on the depth increase parameter H' and maximum Propagation Graph Depth Q_{c-1} in previous round. If neither condition is satisfied, the maximum propagation depth Q_c for that round equals Q_{c-1} . The process of generating the curriculum learning training set is shown in Algorithm 1.

Algorithm 1 The process of generating the curriculum learning training set

Input: original_dataset, current_depth, H' (depth increase parameter), Q_{c-1}

- 1: Initialization: $\text{curriculum_dataset} \leftarrow []$
- 2: **for** each data_item in original_dataset **do**
- 3: $\text{graph_data} \leftarrow \text{data_item}$
- 4: Compute node depths via BFS from source node, $\text{node_depths} \leftarrow \text{compute_node_depths}(\text{graph_data})$
- 5: Compute current allowed max depth Q_c , $Q_c \leftarrow \min(Q_{c-1} + H', \max(\text{node_depths}))$
- 6: **if** $\text{current_depth} > 0$ **then**
- 7: $Q_c \leftarrow \min(\text{current_depth}, Q_c)$
- 8: **end if**
- 9: $\text{valid_nodes} \leftarrow \{v \mid \text{node_depths}[v] \leq Q_c\}$
- 10: $\text{valid_edges} \leftarrow \{e = (u, v) \mid u \in \text{valid_nodes} \text{ and } v \in \text{valid_nodes} \text{ and } e \in \text{graph_data.edges}\}$
- 11: Construct a limited graph with filtered nodes and edges, $\text{limited_graph} \leftarrow \text{build_subgraph}(\text{graph_data}, \text{valid_nodes}, \text{valid_edges})$
- 12: $\text{curriculum_dataset.append}(\text{limited_graph})$
- 13: **end for**
- 14: **return** $\text{curriculum_dataset}$

Output: $\text{curriculum_dataset}$

3.4 Hierarchical Semantic Learning

To better represent historical post information, we employ two encoders to extract features from the same sample data, fusing features from different layers of each encoder via multi-head attention. Contrastive learning is then applied to extract features from different encoder layers and the final fused layer. This module consists of three parts: text feature extraction, attention fusion, and hierarchical contrastive learning.

3.4.1 Text Feature Extraction. We employ two lightweight BERT encoders to perform word-level encoding and post-level encoding on the user's historical post matrix T_i : each encoder consists of two Transformer-based encoder architectures (word-level and post-level). The encoder architecture comprises n stacked encoder layers. Each encoder layer contains two sub-layers: a multi-head self-attention mechanism (MultiHead) and a position-based feed-forward neural network (FFN).

The lightweight BERT encoder generates word-level features w_i^1 through the word-level Transformer, capturing the contextual relationships among words within a single post. First, the initial input $H^{(0)}$ for the word-level Transformer is generated from the

user's historical post matrix T_i :

$$E = \text{Embedding}(T_i) \cdot \sqrt{d},$$

$$H^{(0)} = \text{Dropout}(E + PE),$$

where d is the dimensionality of the word embeddings, and $E \in \mathbb{R}^{r \times s \times d}$ is the word vector matrix obtained by converting T_i into continuous vectors and scaling them. $PE \in \mathbb{R}^{s \times d}$ is the positional encoding matrix. Assume the current l -th layer ($l = 1, \dots, N$) of the word-level Transformer, with input $H^{(l-1)}$ from the previous layer, the intermediate output $Z^{(l)}$ within the l -th encoder layer is generated via the multihead attention mechanism:

$$Z^{(l)} = H^{(l-1)} + \text{Dropout} \left(\text{MultiHead}(\text{LayerNorm}(H^{(l-1)})) \right).$$

Subsequently, the output representation $H^{(l)}$ of the current layer is computed by FFN:

$$H^{(l)} = Z^{(l)} + \text{Dropout} \left(\text{FFN}(\text{LayerNorm}(Z^{(l)})) \right),$$

After stacking through n layers to generate the vector $H^{(n)}$, we apply the layer normalization to derive the contextual representation H_{word} :

$$H_{\text{word}} = \text{LayerNorm}(H^{(n)}).$$

Finally, we use max pooling to generate word-level features w_i^1 :

$$w_i^1 = \max_{q=1}^L (H_{\text{word}}[:, q, :]), \quad (4)$$

where $w_i^1 \in \mathbb{R}^{r \times d}$, r is the number of posts, and L is the length of words in a post. Meanwhile, post vectors p_i^1 is obtained using the above similar transformer architecture.

Another lightweight BERT encoder generates word-level features w_i^2 and post-level features p_i^2 through the same procedure.

3.4.2 Attention Fusion. Similar to [22], we employ an attention mechanism to adaptively learn importance weights for different contextual features. Given that each Transformer layer captures distinct, enhanced semantic information, we integrate contextual representations from all layers to construct the final text representation. First, a linear layer maps the global representations p_i^1 and p_i^2 to the same dimension as w_i^1 and w_i^2 . Then, the word-level features are concatenated with the aligned post-level features:

$$wp_i^1 = \text{Concat}(p_i^1, w_i^1),$$

$$wp_i^2 = \text{Concat}(p_i^2, w_i^2),$$

where wp_i^1 and wp_i^2 are the fused features of the two encoders, respectively, and $\text{Concat}(\cdot)$ is the concatenate function.

The fused features wp_i^1 and wp_i^2 from the two encoders are processed using multi-head attention mechanism to enable interaction between the concatenated sequences. Finally, we employ the average function to obtain the two fused representations M_{atten} and N_{atten} :

$$M_{\text{atten}} = \text{Mean}(\text{MultiHead}(wp_i^1, wp_i^1, wp_i^1)), \quad (5)$$

$$N_{\text{atten}} = \text{Mean}(\text{MultiHead}(wp_i^2, wp_i^2, wp_i^2)), \quad (6)$$

3.4.3 Hierarchical Contrastive Learning. Contrastive learning is to minimize the distance between positive sample pairs and maximize the distance between negative sample pairs, which depends on the design strategy for positive and negative sample pairs. Following the contrastive framework in [5, 9], we adopt normalized temperature-scale cross-entropy loss as the contrastive objective.

The loss function $\mathcal{L}(m'_i, n'_i)$ is defined for a pair (m'_i, n'_i) as follows:

$$\mathcal{L}(m'_i, n'_i) = -\log \frac{\exp(\text{sim}(m'_i, n'_i)/\tau)}{\sum_{k=1}^{2N} \exp(\text{sim}(m'_k, n'_k)/\tau)},$$

where τ is a temperature parameter, k denotes the index of all samples in the batch, $\text{sim}(m', n')$ indicates the cosine similarity between two vectors m' and n' , which are generated by the standard dropout of the Transformer.

To systematically evaluate and enhance the semantic discriminative power of representations across layers, we employ a hierarchical contrastive learning approach to learn precise semantic representations of sentences by computing distances between sample pairs. The objective function is defined as follows:

$$\mathcal{L}_{\text{cl}} = \frac{1}{|B|} \sum_{i \in B} \gamma_i \cdot \mathcal{L}(s_i^m, s_i^n) + \mathcal{L}(s_a^m, s_a^n),$$

where B denotes the selected layer blocks for contrastive learning, γ_i represents the hyperparameters of the i -th layer block, and s_i^m and s_i^n denote the two augmented representations of the i -th layer block, such as the word-level features w_i^1 and w_i^2 , or the post-level features p_i^1 and p_i^2 , respectively. Similarly, s_a^m and s_a^n denote the two augmented representations output by the attention block, such as the attention fusion features M_{atten} and N_{atten} .

3.5 Rumor Classification

Rumor detection is fundamentally a classification task. In this model, we map the learned event-hidden embeddings h_i in Eq.3 through a network architecture comprising a ReLU activation function and a Softmax classification layer to the final classification representation space:

$$\hat{y}_i = \text{Softmax}(\text{ReLU}(W_i h_i + b_i)),$$

where $\hat{y}_i \in \mathbb{R}^{1 \times |C|}$ denotes the probability distribution of the label in all categories, where C is the number of categories. W_i and b_i represent the learnable matrix and bias vector of the fully connected layer, respectively. Next, we apply cross-entropy to obtain the classification loss:

$$\mathcal{L}_{\text{Class}} = - \sum_{i=1}^N \sum_{c=1}^{|C|} y_i^c \log \hat{y}_i^c,$$

where y_i^c denotes the true probability under category c , and N refers to the batch size.

3.6 Loss Function

To enhance model performance, we introduce a semantic alignment loss to align heterogeneous features extracted from rumor propagation graphs and historical propagation tendencies. Its loss function $\mathcal{L}_{\text{align}}$ is defined as follows:

$$\mathcal{L}_{\text{align}} = 1 - \frac{1}{B} \sum_{i=1}^B \frac{h_i^G \cdot h_i^T}{\|h_i^G\| \|h_i^T\|},$$

Table 1: Statistics of the datasets.

Statistic	Twitter15	Twitter16
#Users	276,663	173,487
#Events	1490	818
#true rumors	374	205
#false rumors	370	205
#non-rumors	372	205
#unverified rumors	374	203

where B represents the batch size, the h_i^G denotes the user’s historical preference vector, and the h_i^T denotes the rumor propagation graph vector.

Thus, the total loss $\mathcal{L}_{\text{Total}}$ is expressed as:

$$\mathcal{L}_{\text{Total}} = \lambda_{\text{class}} \mathcal{L}_{\text{Class}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{cl}} \mathcal{L}_{\text{cl}},$$

where λ_{class} , λ_{align} , and λ_{cl} are hyperparameters that balance the three loss functions.

4 Experiment

4.1 Datasets

We conducted experiments on two widely used datasets: Twitter15 [19] and Twitter16 [19]. Table 1 details the specific information for these two datasets. Twitter15 and Twitter16 comprise 1,490 and 818 events, respectively, categorized into four classes: False Rumor (F), Unverified Rumor (U), Non-Rumor (N), and True Rumor (T).

4.2 Experimental Settings

4.2.1 Baselines. We compared our proposed HCSL with baseline methods, as follows:

- BiGCN[3] simultaneously models top-down and bottom-up propagation modes to capture the propagation mechanism.
- CausalRD [39] employs causal graphs and structural negative sampling to detect rumors.
- GACL [26] utilizes propagation structures and adversarial contrastive learning for rumor detection against malicious perturbations and intentional camouflage.
- DDGCN [25] employs a dual dynamic GCN to integrate propagation features and rich semantic knowledge, thereby enhancing the quality of dynamic representations for rumor detection.
- GSMRD [34] employs pivotal semantic graph modeling and semantic-propagation interaction fusion to improve the performance of rumor detection.
- UICL[16] incorporates edge-aware strategies for structural uncertainty and a curriculum learning-guided negative sampling mechanism to improve the performance of rumor detection.
- RAGCL[6] proposes a node-centrality-driven enhancement strategy to focus on critical nodes and deep-level structures.
- AD-GSCL[7] employs propagation structure modeling and cross-domain distribution adaptation to enhance the robustness of rumor detection under conditions of data scarcity and class imbalance.
- DMPRN[36] integrates textual content and propagation structural features to capture the dynamic semantics and temporal dependencies of rumors across multiple granularity levels.

4.2.2 Training Settings. Our model is implemented using PyTorch. Similar to previous studies, we obtained results through cross-validation by dividing the dataset into 5 folds. We set the initial learning rate to $5e-4$, use 200 training epochs, and set the batch size to 128. The graph encoder comprises two layers, with a loss rate of 0.2. For curriculum learning, the initial propagation depth is 1, the depth increase parameter H' is 10, λ_{align} is 0.15, λ_{class} and λ_{cl} are 1, the performance stabilization threshold is $5e-3$, and the performance stagnation patience value is 5.

4.2.3 Evaluation Indicators. To evaluate the model’s performance on the Twitter15 and Twitter16 datasets, we calculated the Acc and F1 scores for all categories. This gives an overall classification performance and discriminative power for each category.

4.3 Experimental Results and Analysis

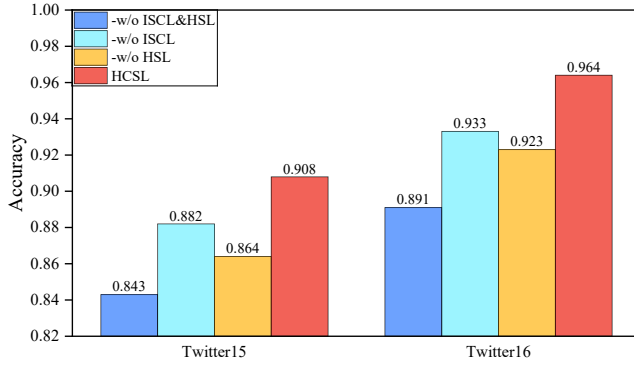
To validate the effectiveness of the proposed model, we conducted comparative experiments against current mainstream rumor detection methods on two real-world social media datasets (Twitter15, Twitter16). The baseline models encompassed diverse technical approaches, including: GCN-based models, such as BiGCN and DDGCN; contrastive learning-based models, such as GACL, UICL, and RAGCL; the causal reasoning-based model CausalRD; the semantic mining-integrated model GSMRD; the real-world scenario-oriented model AD-GSCL, and the latest dynamic architecture model DMPRN. Table 2 presents the performance of each model on both datasets, with the best results highlighted in bold.

The results demonstrate that our proposed HCSL achieves optimal performance across multiple metrics. On both datasets, our accuracy reached its highest levels, improving by 1.1% and 5.0%, respectively, fully validating the overall effectiveness of our proposed intra-sample curriculum learning combined with a hierarchical semantic learning framework. On the Twitter16 dataset, HCSL achieves the highest F1 scores for both the “Non-Rumor (NR)” and “Unverified Rumor (UR)” categories (0.925 and 0.939, respectively). This suggests that higher-quality representations of user historical information are essential for accurately distinguishing authentic information from unverified rumors, thereby providing models with richer contextual evidence. For “False Rumor (FR)” detection, HCSL demonstrated optimal performance on Twitter15. This achievement stems from our model’s effective capture of anomalous patterns in disinformation propagation structures and user behavior. Compared to the traditional GCN models (BiGCN, DDGCN), our framework achieves more robust feature extraction by encoding both rumor propagation graphs and user information. Compared to the contrastive learning models (GACL, UICL, RAGCL), our intra-sample curriculum learning mechanism offers a smarter, adaptive learning paradigm—particularly by avoiding reliance on fixed augmentation rules. This grants greater versatility across complex, dynamic propagation scenarios. Compared to the state-of-the-art models (AD-GSCL, DMPRN), HCSL maintains superior performance across standard benchmarks. This highlights the simplicity and efficiency of our model, achieving top-tier performance without requiring vast amounts of unlabeled data or highly complex architectures.

Based on the above analysis, HCSL demonstrates outstanding performance in overall accuracy and across most categories, fully

Table 2: A Comparative Evaluation of Model Performance on Twitter15 and Twitter16

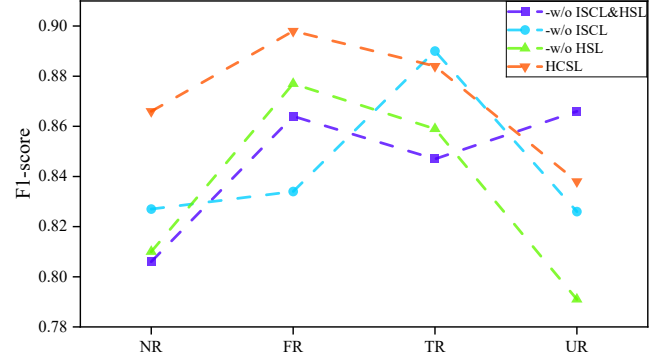
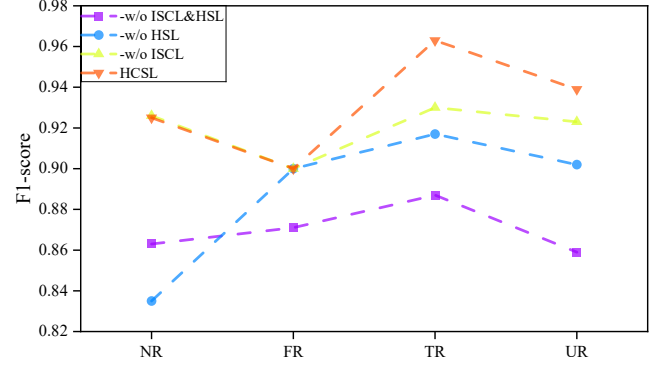
Model	Twitter15					Twitter16				
	Acc	NR(F1)	FR(F1)	TR(F1)	UR(F1)	Acc	NR(F1)	FR(F1)	TR(F1)	UR(F1)
BiGCN	0.886	0.891	0.860	0.930	0.864	0.880	0.847	0.869	0.937	0.865
CausalRD	0.862	0.866	0.861	0.910	0.837	0.891	0.839	0.894	0.939	0.883
GACL	0.882	0.853	0.89	0.902	0.848	0.902	0.912	0.847	0.902	0.876
DDGCN	0.835	0.840	0.850	0.856	0.791	0.893	0.807	0.931	0.946	0.871
GSMRD	0.891	0.884	0.879	0.89	0.885	0.901	0.897	0.892	0.887	0.884
UICL	0.868	0.882	0.878	0.890	0.831	0.907	0.858	0.882	0.934	0.933
RAGCL	0.867	0.891	0.867	0.869	0.835	0.905	0.836	0.923	0.963	0.882
AD-GSCL	0.874	0.886	0.891	0.881	0.836	0.913	0.845	0.926	0.979	0.893
DMPRN	0.897	0.901	0.893	0.931	0.864	0.914	0.873	0.883	0.969	0.931
HCSL	0.908	0.866	0.898	0.884	0.838	0.964	0.925	0.900	0.963	0.939

**Figure 4: Impact of the ablation studies on accuracy for Twitter15 and Twitter16.**

validating the effectiveness of combining ISCL with graph convolutional networks and HSL.

4.4 Ablation Study

To investigate the role of each module in the model, we conducted ablation experiments to compare performance across different module combinations. To evaluate the contributions of the Intra-sample Curriculum Learning (ISCL) and Hierarchical Semantic Learning (HSL) modules, we designed three ablation settings: 1) removing Intra-sample Curriculum Learning (-w/o ISCL); 2) removing Hierarchical Semantic Learning (-w/o HSL); 3) removing both modules simultaneously (-w/o ISCL&HSL). As shown in Figure 4, without ISCL, the accuracy of Twitter15 and Twitter16 decreases by 2.6% and 3.1%, respectively, compared to the original model. As illustrated in Figures 5 and 6, the F1 scores for each category decline to varying degrees, except for the true rumor labels in the Twitter15 dataset, which remain unchanged. This is because ISCL enables the model to start training on simpler samples within the dataset, gradually incorporating more challenging ones. Once this module is removed, the model must immediately process samples of all complexities, making it difficult to capture key propagation patterns. Simultaneously, noise from complex samples interferes with

**Figure 5: Impact of the ablation studies on F1 score for Twitter15.****Figure 6: Impact of the ablation studies on F1 score for Twitter16.**

the feature learning of simpler ones. After removing the HSL module, the accuracy of Twitter15 and Twitter16 decreased by 4.4% and 4.1%, respectively. HSL enables multi-perspective feature fusion and enhances feature representation, serving as the model's core mechanism. After removing both the ISCL and HSL modules, the model's accuracy on the Twitter15 and Twitter16 datasets significantly decreased by 6.5% and 7.3%, respectively. This result validates the

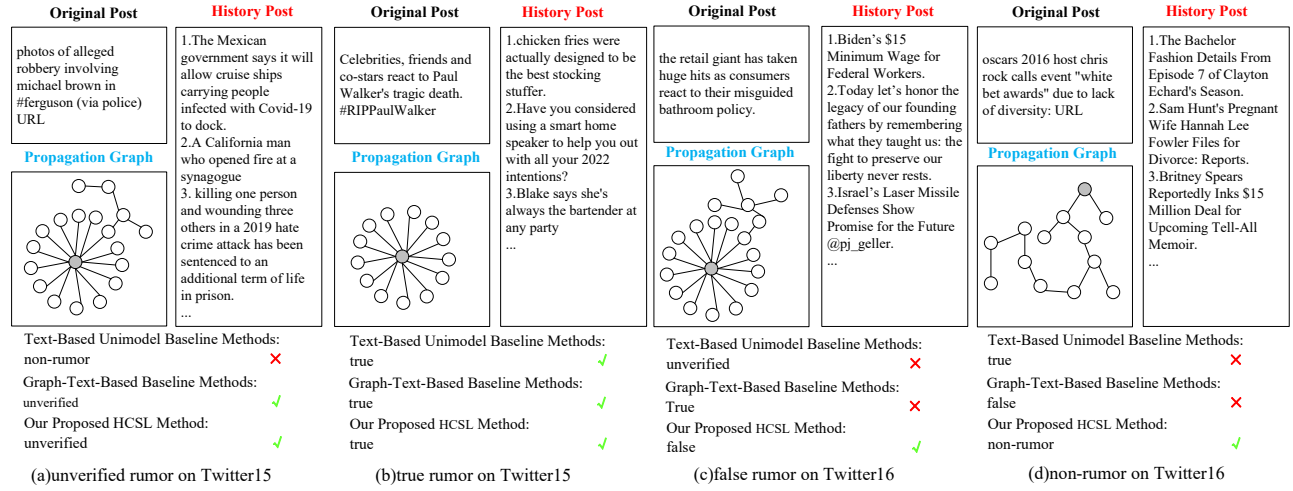


Figure 7: Case studies of four randomly selected rumor events (two from Twitter15, two from Twitter16). Propagation graphs are shown, with gray nodes indicating source posts and other nodes representing comments or retweets.

critical role of our proposed ISCL and HSL modules in enhancing model performance.

5 Case Study

To better demonstrate the strengths of our model, we randomly selected two data points from each of the Twitter15 and Twitter16 datasets. Figure 7(a) shows data from Twitter15, with the text “photos of alleged robbery involving michael brown in #ferguson (via police) URL”. This post received 167 retweets and 70 replies; Figure 7(b) shows an event from Twitter15: “Celebrities, friends and co-stars react to Paul Walker’s tragic death. #RIPPaulWalker”. This original post received 180 retweets and 14 replies. Figure 7(c) shows an event from Twitter16: “The retail giant has taken huge hits as consumers react to their misguided bathroom policy”. This original post was retweeted 65 times and received 10 comments. Figure 7(d) shows data from Twitter16: “Oscars 2016 host Chris Rock calls event ‘white bet awards’ due to lack of diversity: URL”. This original post was retweeted 217 times and received 36 comments. These events were detected using text-based modeling, propagation graph methods, and our model, respectively. Based on the results in Figure 7, we can observe that: (a) as the propagation graph deepens, it provides more information, leading to successful predictions for all propagation graph-related methods; (b) since the propagation graph is relatively shallow, its impact on the model is minimal, resulting in successful predictions for all three methods; (c) Although the propagation graph structure differs little from (a), our model successfully predicted only because it generates higher-quality textual features from users’ historical post information. (d) Due to the propagation graph’s great depth and complex structure, traditional GCN processing introduces substantial noise, resulting in prediction failures. Our model, however, successfully predicted this event by performing a gradual, in-depth detection of the propagation graph, thereby reducing noise interference.

6 Conclusion and Future Work

In this paper, we propose HCSSL, a model primarily composed of two components: ISCL and HSL. ISCL reduces noise interference by mimicking human learning processes and designing an intra-sample curriculum learning strategy. HSL generates high-quality text representations by contrastively learning multi-level features through the collaborative interaction of dual encoders. We conducted extensive experiments on two public datasets, demonstrating that our model outperforms state-of-the-art approaches. Future work will focus on integrating large language models and multimodal learning to enhance text representation and rumor detection capabilities.

Acknowledgments

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A Related Work

A.1 Rumor Detection Methods

Early rumor detection methods primarily focused on crafting manual features and integrating them with traditional classification algorithms for identification [28]. Nevertheless, as society has been evolving ceaselessly, there has been an increasing demand for the efficiency and accuracy of information classification. This has rendered traditional manual feature extraction methods insufficient to address this change. In recent years, rumor detection based on deep learning has become a research highlight. This is because it can automatically extract features and efficiently identify rumors [13, 32]. Based on the data they employ, existing approaches can be grouped into three categories: content-based methods, propagation-structure-based approaches and multi-source information fusion methods. Content-based methods mainly depend on discerning the lexical, syntactic, and thematic traits inherent to the sentence itself. Propagation-structure-based approaches focus on analyzing the diffusion paths of comments, shares, and retweets related to rumors. Multi-source information fusion methods combine post content, user attributes, historical behavior, and the intricate interrelationships among these elements to detect rumors.

In recent years, the interactive features of tweets—such as comments, retweets, and likes—have been recognized as indicators of the breadth and influence of information dissemination. This is because increased retweets, comments, and likes directly expand a tweet’s reach and visibility [45]. These metrics effectively track the propagation pathways of rumors. In related work, ESODe [20] used the number of comments and retweets as input features. Others, in turn, Qourbani et al. [24] expanded the feature set by including additional details such as the number of likes, the users who commented, those who retweeted, and the tweet’s posting time. To more precisely represent the propagation patterns of rumors across various time stages, CUSMAXLSTM [17] constructed a propagation tree structure and analyzed users’ retweet behavior patterns to extract the dynamic alterations in propagation features. ACMR [29] generates representations that exhibit both modal invariance and semantic discriminability by integrating modal adversarial constraints and triplet structure preservation loss within a shared subspace. PPA-WAE [37] first established a basic propagation network structure and then introduced two learnable parameters to more precisely model the differences in interaction patterns between nodes and their parent or child nodes. BAET [38] significantly enhances the accuracy of social media rumor detection by jointly modeling post content, author characteristics, and their propagation structures, while incorporating root-aware and tree-aware attention mechanisms. Zhong et al. [42] note that existing studies often concentrate on propagation analysis while overlooking the innate clustering structure of rumors. This oversight leads to sub-optimal detection performance during the early or incomplete propagation phases. To tackle this, their study combines aggregation characteristics,

textual features, and propagation features to boost detection performance. Recent research [8, 23, 35] has shown that integrating social context and source-attribute information can substantially improve rumor detection performance. For example, HPCL [41] effectively improved rumor detection by simultaneously leveraging both types of features.

A.2 Curriculum Learning

Curriculum Learning was first proposed by Bengio et al. [2] to simulate the human learning process from easy to difficult tasks. This method improves optimization and generalization performance by designing difficulty metrics for data samples, arranging them in ascending order of difficulty, and organizing the model training process accordingly. As a general training strategy, curriculum learning has been widely applied across multiple research domains. Studies in computer vision [10], natural language processing [27], and reinforcement learning [21] have validated Curriculum Learning’s effectiveness in accelerating convergence and enhancing model robustness.

Curriculum learning is applied to scenarios such as noisy labeled data and social media, aiming to mitigate the negative impacts of noisy samples [41]. However, most of these methods focus only on sorting difficulty across samples, without modeling structural differences within individual samples. As a result, they remain inadequate for social media data, which exhibits complex propagation structures and contains multi-level interaction information. EfficientTrain++ [31] proposes a concise and broadly applicable efficient learning paradigm. It transforms the traditional approach of arranging course materials by the difficulty level of each sample into measuring the “simpler” and “more complex” components of each sample, progressively introducing the more complex aspects during model training. This approach not only helps reduce noise interference but also enhances the extraction and utilization of key structural information within each propagation graph. In comparison, the latter training strategy proves more effective. The proposed intra-sample curriculum learning strategy extends curriculum learning from “inter-sample” to “intra-sample”. By fine-grained modeling of propagation graph structures, constructing sample-based difficulty metrics based on structural depth, and adaptively adjusting the model’s focus on sub-structures during training, it more effectively suppresses noise interference and extracts valuable information.

A.3 Contrastive Learning

Contrastive learning aims to learn representations by mapping similar samples closer together and dissimilar samples farther apart [11]. Driven by growing demand for feature representations, contrastive learning has advanced rapidly in natural language processing and has been widely applied across various tasks. For instance, to address the collapse of BERT-derived representations and their limited transferability to downstream tasks, ConSERT [33] proposed a contrastive learning-based framework for optimizing sentence representations. MMEN [12], which leverages multimodal augmentation mechanisms and an adaptive contrastive learning module, significantly mitigates the tendency to neglect other modalities when text-based modalities dominate. AFdgc [44] combined Transformers

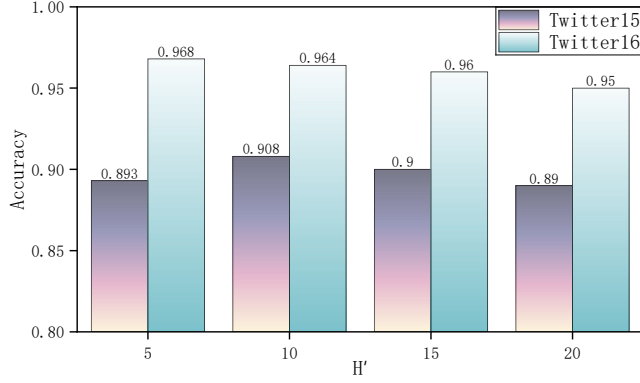


Figure 8: Results on Twitter15 and Twitter16 with respect to the depth increase parameter H' .

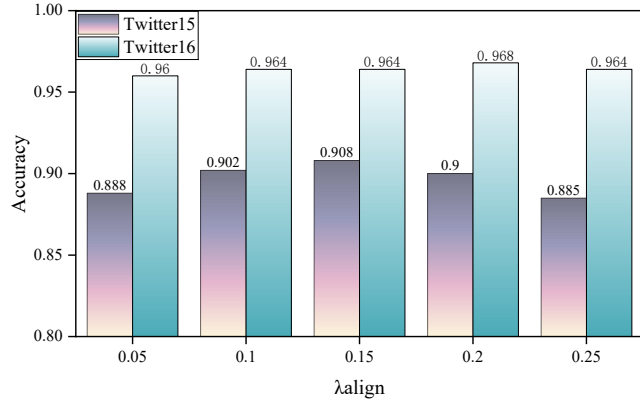


Figure 9: Results on Twitter15 and Twitter16 with respect to the semantic alignment parameter λ_{align} .

with unsupervised contrastive learning to resolve temporal consistency and data integrity issues in dynamic graph representation learning. CLBERT [4] employed contrastive learning to enhance

representation capabilities by comparing layer-by-layer representations, aiming to generate superior semantic representations for BERT sentences. Inspired by these works, we integrate contrastive learning with Transformers from different layers of a lightweight BERT model to enhance sentence representations.

B Parameter Sensitivity Analysis

In this section, we evaluate the impact of the depth increase parameter and the semantic alignment parameter. All other parameters were kept fixed in the training configurations.

B.1 Effect of Depth Increase Parameter

With the other parameters unchanged, we conducted a sensitivity analysis of the depth increase parameter H' for the depth of the propagation structure map in the intra-sample curriculum learning. The main intervals are 5, 10, 15, and 20. The influence of this parameter on accuracy is assessed by increasing the depth in 5-unit increments. Figure 8 shows that Twitter16 gradually decreases with depth, whereas Twitter15 reaches a peak at depth 10. Although Twitter16 achieves the highest accuracy at a graph depth of 5, Twitter15 has relatively low accuracy. When the depth is 10, the model performs best. Overall, the accuracy of HCSL decreases with depth on both datasets. The main reason is that as the depth H' increases, the number of forwarding nodes increases with their distance from the root node.

B.2 Effect of Semantic Alignment Parameter

We set the semantic alignment parameter $\lambda_{align} \in \{0.05, 0.1, 0.15, 0.2, 0.25\}$ and tested the impact of this parameter on the model within this range. As shown in Figure 9, the model achieves its best performance on Twitter15 when λ_{align} is set to 0.15, while it performs best on Twitter16 with $\lambda_{align} = 0.2$. Overall, the model demonstrates optimal performance when the parameter is set to 0.15. Twitter15 exhibits greater sensitivity to the semantic alignment loss parameter, primarily due to significant differences in its user features and structural characteristics, thus requiring stronger semantic alignment to mitigate feature inconsistencies.